Mentor briefing | Seva First Innovation Challenge

AgriGuard

Early Crop Disease Screening and Safe Advisory for Small Farmers

A phone photo of a leaf → healthy or likely disease, how serious, and safe next steps in English or Tamil — or "Not sure, ask your KVK".

How we will present (5 speakers, about 5 minutes)

#	Topic	Time	Speaker name
1	The Problem	1 min	
2	Our Solution – what the farmer does	1 min	
3	How the AI works and how accurate it is	1 min	
4	Technology & system design	1 min	
5	Impact, cost, next steps & submission	1 min	

How to use this PDF: each member picks one topic, writes their name above, and practises their page. Read the "What to say" box in your own words — don't memorise word for word. Everyone should know the numbers on the last page.

Our one-line pitch (everyone should know this)

"AgriGuard lets a farmer photograph a sick leaf and, within seconds, know if it is healthy or diseased, how serious it is and what safe step to take next — and when to call an expert."

Theme / Sub-theme	Samriddh Annadata, Samriddh Bharat · Agricultural Innovation
Track / Stage	Track 1: Product / Technology · Prototype
Deadline	24 October 2026 (SFIC portal)

1 The Problem

Speaker 1: _____ · about 1 minute

Goal: Make the mentor feel why early crop-disease detection matters for Indian farmers.

WHAT TO SAY (READ ALOUD, IN YOUR OWN WORDS)

"Good morning. Our project is **AgriGuard**. Crop diseases and pests are one of the biggest reasons Indian farmers lose income. ICAR has said about **30–35% of crop yield** is lost to pests every year."

"The real problem is **late detection**. Most small farmers identify a disease by eye or ask the pesticide shop. There is less than **one extension worker for every 5,000 farmers**, so expert help comes late. Farmers then spray several chemicals by guesswork."

"For example, in 2023 pink bollworm destroyed about 65% of cotton in Punjab and Haryana. In Telangana, a chilli farmer said she used six pesticides and none worked. In Tamil Nadu's Pollachi belt, coconut yields fell 30–40%. The pattern is always the same: the problem is found late, and relief comes after the crop is lost."

Key points to remember

- Problem statement: small farmers cannot identify crop diseases early or get timely advice, so outbreaks destroy their harvest and income.
- 30–35% of yield lost to pests (ICAR). Extension worker ratio below 1:5,000 (norm 1:750–1:1,100).
- Real cases: cotton 2023 (Punjab/Haryana/Rajasthan), chilli 2021–22 (Telangana/AP), coconut 2024–25 (Tamil Nadu), banana 2020 (Bihar/UP), soybean 2025 (Madhya Pradesh).
- Theme: Samriddh Annadata, Samriddh Bharat → Agricultural Innovation.

Mentor may ask

Q: Where did you get these numbers?
A: From published sources: ICAR statement (FreshPlaza, 2017), NAAS/ICRISAT 2025 for the extension ratio, and Down To Earth, DT Next and The Quint news reports. All are listed in our case-study document.

Q: Who exactly is the user?
A: Small and medium farmers with a basic smartphone, plus extension officers and KVK staff who support them.

Q: Is this a real problem in Tamil Nadu?
A: Yes. Coconut root wilt and whitefly cut yields 30–40% in Pollachi in 2024–25, and 1.9 lakh acres were reported affected.

2 Our Solution – What the Farmer Does

Speaker 2: _____ · about 1 minute

Goal: Explain AgriGuard simply, from the farmer's side, and show the flow.

WHAT TO SAY (READ ALOUD, IN YOUR OWN WORDS)

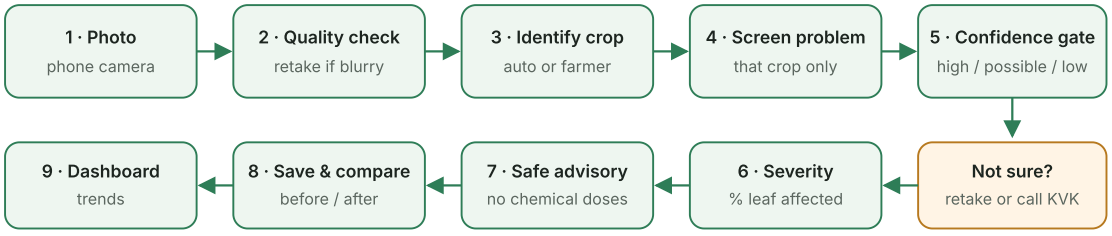
"AgriGuard is a mobile web app. The farmer opens it in the phone browser — no app install. He takes a photo of one affected leaf. In a few seconds, AgriGuard tells him whether the leaf looks **healthy** or shows a **likely disease**, how confident it is, how much of the leaf is affected, and what to do next — in **English or Tamil**."

"Two things make it safe. First, if the photo is unclear or the AI is not sure, it says **'Not sure'** and tells the farmer to retake the photo or contact the KVK — it never forces a wrong answer. Second, it **never gives pesticide brand names or doses**; it gives safe steps like removing infected leaves, and sends the farmer to the agriculture officer for treatment."

Key points to remember

- Steps: open app → choose crop (or auto-detect) → photo → result → advice → saved in history.
- Result shows: healthy / possible disease / not sure, confidence %, severity (Low / Moderate / High), early-signs warning.
- History and before/after comparison show if the problem is spreading.
- English + Tamil; optional ESP32 sensor kit shows soil moisture, temperature and humidity.

Flow to show



Optional field sensing

ESP32 kit with soil-moisture, temperature and humidity sensors sends readings every minute.
Shown as context (e.g. humid, wet weather favours fungal disease), never as a diagnosis.

Mentor may ask

Q: Why not tell the farmer which pesticide to use?
A: Wrong advice can harm the crop, the farmer's health and the soil. Treatment decisions must come from local experts, so every advisory ends with a KVK referral. This rule is checked by an automated test.

Q: What if the farmer has no internet?
A: Today it needs internet. Our roadmap includes an offline mode, because the model is small (10 MB) and can run on the phone.

Q: Can farmers who can't read use it?
A: The design uses icons and Tamil; voice output is planned.

3 How the AI Works – and How Accurate It Is

Speaker 3: _____ · about 1 minute

Goal: Explain the AI model honestly: what it was trained on, what it can do, and its limits.

WHAT TO SAY (READ ALOUD, IN YOUR OWN WORDS)

"The AI is an image model called **MobileNetV2**, which is small enough to run in a phone browser. We trained it on about **9,200 lab leaf photos** from PlantVillage and **2,300 real field photos** from PlantDoc. It covers **5 crops — tomato, potato, chilli/pepper, maize and grape — and 23 classes**."

"We tested it on photos it had never seen. On lab photos it names the right disease **94.7%** of the time. On real field photos it is **61.5%**, because field photos have soil, shadows and other leaves. But for the most important question — is the leaf healthy or not — it is much better: **24 of 28 healthy field leaves** were reported healthy, and **105 of 120 diseased leaves** were flagged as a problem."

"We also added a **leaf-only colour check** that ignores the background, so soil or a dark background is never called a disease."

Key points to remember

- Model: MobileNetV2 + our trained classifier head; 10 MB; same file runs on laptop server and in the browser.
- Data: PlantVillage (CC BY-SA 3.0) + PlantDoc field photos (CC BY 4.0); field test set kept separate.
- Confidence rules: ≥80% → result; 55–80% → 'possible'; below → 'Not sure'.
- For crops not yet covered (rice, banana, cotton…): no disease is guessed — only 'no visible problem' or 'not sure'.

94.7%	61.5%	24 / 28	105 / 120
correct disease – lab photos (2,300)	correct disease – field photos (148)	healthy field leaves reported healthy	diseased field leaves flagged

Mentor may ask

Q: Why is field accuracy lower?
A: Lab photos have plain backgrounds; real fields have soil, shadows and overlapping leaves. That is why our next step is collecting 30–50 real photos per disease from local farms and fine-tuning.

Q: Did you build the model yourself?
A: We used a pretrained backbone (transfer learning) and trained our own classifier on crop-disease datasets. This is the standard approach and lets us focus on safety and the farmer experience.

Q: What happens if the AI is wrong?
A: It shows its confidence, says 'Not sure' when unsure, and every result recommends confirming with the KVK. It is a screening tool, not a final diagnosis.

4 Technology & System Design

Speaker 4: _____ · about 1 minute

Goal: Show the mentor that the system is really built and how the parts connect.

WHAT TO SAY (READ ALOUD, IN YOUR OWN WORDS)

"The system has four parts. The **frontend** is a Next.js web app that works on any phone browser, with English and Tamil. The **backend** is Python FastAPI, which checks photo quality, runs the AI model, estimates severity and saves results. The **database** stores accounts, scans and sensor readings, with each farmer's photos kept private. And an optional **ESP32 kit** sends soil moisture, temperature and humidity every minute."

"The AI model can run in two places: on the laptop server, or directly inside the phone browser. So the website gives the same result even without our server. We have **40 automated tests** that check photo quality, confidence rules, severity, the no-chemical rule and the real model."

Key points to remember

- Frontend: Next.js PWA, Tailwind, English/Tamil, history, compare, dashboard.
- Backend: FastAPI, photo quality check, confidence policy, severity, advisory stored as editable data.
- AI: ONNX model (onnxruntime on server, onnxruntime-web in browser).
- Data: SQLite on laptop / Supabase (PostgreSQL) in cloud. Hardware: ESP32 + soil moisture sensor + humidity sensor + LCD display (≈ ₹1,500).
- Testing: 40 automated backend tests pass.

Architecture to draw on the board

Phone browser →	Backend (FastAPI) →	Database
Photo, crop, language; AI can also run here	Quality check → AI model → confidence → severity → advice	Accounts, scans, history, sensor data
ESP32 sensor kit → backend every 60 s (optional)		

Mentor may ask

Q: Why a web app and not an Android app?
A: Farmers don't need to install anything; it works on low-end phones through the browser. We also have an Android wrapper app.

Q: Is the ESP32 required?
A: No, it is optional. It adds field context (humid, wet weather favours fungal disease), never a diagnosis.

Q: How do you protect farmers' data?
A: Each user has a private account and photo folder. Before public launch we are also fixing login security (removing test shortcuts and properly hashing passwords).

5 Impact, Cost, Next Steps & Submission

Speaker 5: _____ · about 1 minute

Goal: Close strongly: who benefits, how cheap it is, what we will do next, and what we need from the mentor.

WHAT TO SAY (READ ALOUD, IN YOUR OWN WORDS)

"AgriGuard helps farmers act **early**, protecting yield and cutting money wasted on wrong sprays. It reduces unnecessary chemicals, and it helps KVKs reach many more farmers."

"It is low-cost: the whole working product costs about **₹1,500** — ESP32, soil moisture sensor, humidity sensor and LCD display — and the software is free because we used open-source tools. The app is **free for farmers**. Our next steps are: collect real leaf photos from local farms to improve accuracy, add more crops like rice, chilli, brinjal and banana, have experts review the advice, and run a **pilot with 50–100 farmers through a KVK**."

"We are submitting to the **Seva First Innovation Challenge** under Samriddh Annadata, Samriddh Bharat, Track 1, at Prototype stage. Sir/Madam, we would like your guidance on the pilot and on connecting with a KVK or agricultural expert."

Key points to remember

- Impact: economic (yield, input cost), social (same-day guidance, Tamil), environmental (fewer chemicals), institutional (KVK reach, early outbreak alerts).
- Cost: product ≈ ₹1,500 per kit (ESP32 ₹600, soil moisture ₹200, LCD ₹200, humidity ₹150, wires/soldering/packing ₹350); software ₹0; free for farmers.
- Adopters: KVKs, State Agriculture Department, TNAU / ICAR institutes, FPOs, NABARD, CSR foundations.
- Submission: SFIC portal, deadline 24 Oct 2026 — concept PDF and portal answers ready.

Mentor may ask

Q: Why is it so cheap?
A: The hardware is only about ₹1,500 and all software is open-source. The AI runs on the farmer's own phone, so there is no expensive server.

Q: How will you make money / sustain it?
A: Free for farmers; kits and hosting can be funded by agriculture departments, KVKs, FPOs or CSR partners at scale.

Q: What support do you need?
A: Mentorship from a plant pathologist, access to labelled field images, cloud credits, seed funding of about ₹20,000–50,000, and a KVK or FPO pilot partner.

Q: What is your current stage?
A: Prototype: the full app works end to end with a trained model; next is field validation and a pilot.

Numbers & answers everyone should know

Fact	Value
Crop yield lost to pests in India	30–35% (ICAR)
Extension workers to farmers	below 1 : 5,000 (norm 1:750–1:1,100)
Crops covered by the AI model	5 – tomato, potato, chilli/pepper, maize, grape (23 classes)
Training photos	~9,200 lab (PlantVillage) + ~2,300 field (PlantDoc)
Accuracy – lab / field	94.7% / 61.5% correct disease name
Healthy field leaves reported healthy	24 of 28
Diseased field leaves flagged	105 of 120
Model size	10 MB (runs in phone browser)
Automated tests passing	40
Product cost (one field kit)	≈ ₹1,500 – ESP32 ₹600, soil moisture ₹200, LCD ₹200, humidity ₹150, wires/soldering/packing ₹350
Software cost	₹0 (open-source, free hosting tiers); app free for farmers
Pilot plan	50–100 farmers through a KVK or FPO

Three rules that make AgriGuard safe

1. **Says "Not sure"** when the photo is unclear or confidence is low.
2. **No pesticide names or doses** — always refers treatment to the KVK / agriculture officer.
3. **Only checks the leaf** — soil, pots and dark backgrounds are never called disease.

Honest limits (say these if asked — honesty impresses mentors)

- Field accuracy for the exact disease name is about 6 in 10; healthy-vs-problem is much better.
- Only 5 crops have the full AI model today; others get a basic leaf check.
- Needs internet today; offline mode is planned.
- Not yet tested with real farmers — that is the pilot.

Ask the mentor

- Can you connect us with a KVK, TNAU or agriculture expert to review our advice and run a pilot?
- Is Prototype the right stage to select, and is our track choice correct?
- What should we strengthen in the concept document before 24 October?

Sources: ICAR statement via FreshPlaza (2017); ICRISAT/NAAS (2025); Down To Earth (2020, 2022, 2023); DT Next (2024); The Quint (2025); Global Agriculture and LatestLY (2025). Accuracy measured on held-out PlantVillage and PlantDoc test photos, 13 Oct 2026.